

Impact of COVID-19 Lockdowns on Air Quality in Bangladesh: Analysis and AQI Forecasting with Support Vector Regression

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Abstract—Over the past few decades, air pollution has emerged as a significant environmental hazard, causing premature deaths in Southeast Asia. The proliferation of industrialization and deforestation has resulted in an alarming increase in pollution levels. However, the COVID-19 pandemic has significantly reduced the amount of volatile organic compounds and toxic gases in the air due to the decrease in human activity caused by lockdowns and restrictions. This study aims to investigate the air quality in various geographical areas of Bangladesh, comparing the air quality index (AQI) during different lockdown periods to equivalent eight-year time spans in 10 of the country's busiest cities. This study demonstrates a strong correlation between the rapid and widespread dispersion of COVID-19 and air pollution reduction in Bangladesh. In addition, we evaluated the performance of Support Vector Regression (SVR) in AQI forecasting using the time series dataset. The results can help improve machine learning and deep learning models for accurate AQI forecasting. This study contributes to developing effective policies and strategies for reducing air pollution in Bangladesh and other countries facing similar challenges.

Index Terms—COVID-19, air pollution, air quality index (AQI), time series analysis, Support Vector Regression (SVR), machine learning, environmental risk.

I. INTRODUCTION

Air pollution is a major problem that severely affects both urban and rural areas' environmental quality [1]. Big cities around the globe, specifically cities in developing countries, are severely affected by air pollution. Out of the 6,475 cities examined, only 222 countries had average air quality that matched WHO standards [2]. The main contributors to air pollution are motor vehicles, brick kilns, diesel generators, and industries. The pollutant species are carbon monoxide (CO), hydrocarbons (HC), nitrogen oxides (NOx), etc and these pollutants are recognized as one of the main detriments to human health [3], [4]. According to the World Health Organization, 7 million people are at health risk due to

air pollution [5]. Thus, a continuous monitoring system and analysis of the air quality index (AQI) is essential to keep the pollution level low so that it would not exceed the safe limit. The air quality index (AQI) is an index for reporting air quality daily.

Our study aims to investigate the Air Quality Index (AQI) trends of Bangladesh, which is known to have the most polluted air in the world, over the past few years, and examine how the country's air quality changed during the COVID-19 pandemic [6]. We found that the concentration of AQI is alarmingly high all over the country, particularly in major cities such as the capital Dhaka, Chittagong, and Gazipur. Dhaka is frequently named the most polluted city, with the poorest Air Quality Index [7]. Despite the fact that the country's AQI is substantially higher than the global average, our study discovered that during the COVID-19 lockdown, the AQI significantly decreased. The highest AQI measured in the country during the 2020 lockdown was much lower than the highest AQIs observed in 2019 or 2021.

Additionally, the analysis of the data we extracted shows that the lockdown period in 2020 had the lowest air quality ratings compared to the lowest values reported in 2019 and 2021. Our dataset contains AQI data from February 17, 2014, through July 6, 2022. The dataset had approximately 26000 values and five columns: Date, Location, Air Quality Index (AQI), and AQI Category & Range. The following are the contributions of our work:

- In order to analyze the changes in air quality during the COVID-19 pandemic, we extracted data from the CASE website using web scraping techniques [8]. Initially, the dataset had several missing values and was not sorted, but with the help of our script, we collected the dates of each data and sorted the dataset for convenient time-series

analysis.

- To gain a better understanding of the changes in air quality index (AQI) over time, we leveraged the COVID-19 outbreak as a natural phenomenon and divided the dataset into three distinct periods: pre-COVID-19 (2014-2019), COVID-19 lockdown (2020), and post-lockdown (2021).
- We evaluated the performance of Support Vector Regression (SVR) on our dataset using Mean Absolute Percentage Error (MAPE) for time-series forecasting. Our study demonstrates the potential of using machine learning and deep learning models to address air pollution issues in Bangladesh, which may encourage further research in this area to support mitigation plans against air pollution.

II. RELATED WORKS

Numerous research has been conducted to determine the impact of how the COVID-19 lockdown has affected air quality and pollution all over the world. In cities of China, a drop in daily NO_2 concentrations of 3% to 58% was noted during the lockdown [1]. Over some of India's major cities like Delhi, Mumbai, Hyderabad, Kolkata, and Chennai, $\text{PM}_{2.5}$ and AQI kept falling and a declining trend in tropospheric NO_2 concentration was seen during the lockdown period in [9]. Some studies have also been conducted in Bangladesh. It was found that in April and May 2020, the concentrations of $\text{PM}_{2.5}$, NO_2 , SO_2 , and CO decreased on average by 30% during the lockdown compared to the previous year, while the AQI decreased on average by roughly 35% during the lockdown period [10]. In [11] authors also found the same, including about 10% reduction in O_3 concentration. Authors in [5] found that the air quality index (AQI) had a strong negative correlation with cumulative cases of COVID-19 in Dhaka city. A similar trend was also found in Chittagong city, the main port city of the country. The study found AQI got improved up to 26% during the COVID lockdown phases compared to previous years [12]. Furthermore, authors in [13] found that along with decreasing NO_2 and $\text{PM}_{2.5}$ concentrations, the Aerosol Optical Depth (AOD) also experienced a considerable reduction up to 47% in all major cities. So far, a study has yet to be done which gives a complete overview of the trend of AQI over the major cities of the country. The main objective of our study was to explore the trends of the Air Quality Index (AQI) of 10 major locations in Bangladesh over the last several years and investigate the changes in the Air Quality of the country during the COVID lockdown.

III. DATASET DESCRIPTION

A. Study Area

In our research, we focused on 10 cities (Barishal, Chittagong, Cumilla, Dhaka, Gazipur, Khulna, Mymensingh, Narsingdi, Rajshahi, and Sylhet) and identified various sources of air pollution, including high-sulfur fuel-powered cars, unregulated brick kilns, construction activities, and industrial operations, among others. Recent research has suggested that lockdowns, which result in reduced activity and emissions

from sources such as road traffic and industries, have contributed to improvements in air quality.

B. Data Source and Pre-processing

The data was collected from the Clean Air and Sustainable Environment (CASE) project under the Ministry of Environment and Forest of the Government of Bangladesh [8].

The AQI data was categorized from good to extremely unhealthy as recommended by the Ministry of Environment and Forests, Government of the People's Republic of Bangladesh shown in Table I. The dataset contained the AQI of several locations in the study area for a period from 2014 to the most recent time of data download. We extracted data from the 17th of February 2014 to the 6th of July 2022. The dataset contained five columns: Date, Location, Air Quality Index (AQI), AQI Category, and Range, and it had about 26000 values. The dataset had AQI values for 10 significant locations in Bangladesh, namely Barishal, Chittagong, Comilla, Dhaka, Gazipur, Khulna, Mymensingh, Narayanganj, Narsingdi, Rajshahi, and Sylhet. We sorted the data that we collected from the website. Upon further investigation, it was also found that there were 13 NaN values in the Date column, 49 NaN values in the Location column, 641 NaN values in the AQI column, and 63 NaN values in the AQI Category column. For our convenience, we decided to drop the NaN values. There were also about 5000 instances in the AQI column where AQI values were DNA (missing). It was replaced with NaN and then dropped from the dataset. The AQI Range column was insignificant for the study, as our principal analysis would consist of the AQI values, date, and location columns. Hence, the range column was not considered and was dropped from the dataset. Some of the spelling errors in the dataset were fixed, and after that, the dataset was sorted date-wise, making time series analysis more convenient.

For the prediction model, we used Support Vector Regression (SVR) [14]. As SVR is based on the structural risk minimization principle, it is less prone to overfitting and tries to find a balance between model complexity and predictive accuracy [15], [16]. This supervised learning algorithm can handle continuous and categorical data, and noise and outliers are also well handled.

Most linear models are not well-suited to capture the non-linear patterns in time-series data. In this case, SVR can process non-linearity in time-series forecasting and produce successful predictions. SVR tries to fit a hyperplane that best separates the data points while maximizing the margin between them, and it does not depend on the dimensionality of the input space.

IV. METHODOLOGY

We have divided our work into two phases: data pre-processing and cleaning for comparative analysis in phase one, and implementing and evaluating the performance of SVR on this dataset in phase two. Fig. 1 presents the methodology employed in this study.

We used the dataset from the Clean Air & Sustainable Environment (CASE) project website [8]. It includes AQI data

TABLE I
APPROVED AIR QUALITY INDEX (AQI) FOR BANGLADESH

Air Quality Index	Air Pollution Level	Health Implications
0-50	Good	Air quality is considered satisfactory with little or no risk of air pollution.
51-100	Moderate	Air quality is acceptable though pollution level makes concern for a group of people who are sensitive to air pollution.
101-150	Caution	Members of sensitive groups may have health effects, and the general public is not likely to be affected.
151-200	Unhealthy	Everyone may experience health hazards, and sensitive people may experience more serious health effects.
200-300	Very Unhealthy	The entire population may experience more serious health effects.
301-500	Extremely unhealthy	Everyone is likely to be affected by more serious health effects.

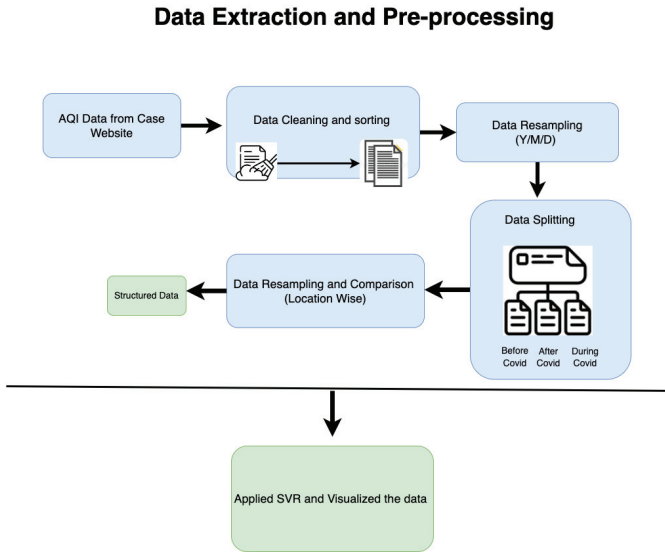


Fig. 1. Methodology employed in this work

from February 17, 2014, through June 6, 2022. We split the dataset into three distinct time periods based on the COVID-19 lockdown, namely: Before COVID-19 (2014-2019), During lockdown (2020), and After lockdown (2021)

As Bangladesh got its first three COVID-19 cases in March 2020, the before-COVID-19 period was taken from the earliest data we have in our dataset, 17 February 2014 to 31 December 2019. During the lockdown, the time period was taken from 1 January 2020 to 31 December 2020. The lockdown adhered better to the first year, so we considered the entire year to be under lockdown. The period was taken from 1 January instead of 8 March (the date of the first case) only for generalization purposes. After the lockdown time period was taken from the following year, 1 January 2021, to 31 December 2021, as the lockdown was significantly reduced, all the mills, factories, and offices in the country were running in full swing, which was not the case in the previous year. We calculated the maximum AQI, minimum AQI, mean AQI, and standard deviation for each period to better understand the distribution of AQI values and identify any significant trends or changes over time. We resampled the data at different time intervals to analyze the AQI patterns over different periods and to facilitate comparison across different locations. It was resampled monthly, yearly, and later by location to analyze the AQI patterns of different locations over different periods.

After analyzing the AQI patterns of different locations, the

dataset was split into training and testing sets. The training set consisted of the data from 2014-02-28 to 2020-12-31, and the testing dataset consisted of the data from 2021-01-01 to 2022-06-01. The AQImean column was scaled to be in the range (0, 1) using MinMaxScaler in order to generalize the data. Following the preprocessing, SVR was implemented using RBF kernel with the hyperparameters gamma, C, and epsilon set as 0.5, 10, and 0.05, respectively, and the model was fitted using the training data. For our model evaluation, we scale back the data to our original scale using inverse transformation. We evaluated our model performance by comparing the original and predicted AQI mean plot and calculating the MAPE (Mean Absolute Percentage Error) on the dataset.

V. RESULT ANALYSIS AND DISCUSSION

We have divided this section into three parts. Part V-A contains the model analysis, part V-B contains the location-wise analysis, and part V-C discusses the overall analysis.

TABLE II
ACTUAL VS PREDICTED MEAN AIR QUALITY INDEX (AQI)

Date	Actual AQI	Predicted AQI
2014-09-02	46.5	46.5
2014-09-03	48.875	48.875
2015-01-01	230.75	230.75
2015-01-02	185.25	185.12352416
2015-01-03	174.25	174.07473857
2015-01-07	210	210.08719749
2015-01-11	175.375	175.20197959

A. Model Analysis

In order to evaluate our model, we used the Mean Absolute Percentage Error (MAPE) calculation method. The MAPE is obtained by dividing the sum of absolute forecast errors by the actual values for each period. Typically, a MAPE of less than 5% is considered perfectly accurate, while a MAPE between 10% - 25% is considered to have low but acceptable accuracy. Our model had a MAPE of approximately 6.2%, which is considered good and acceptable. The SVR model performed very well, predicting accurately the AQI values. Fig. 2 shows a plot comparing the predicted values with the actual values, and we can observe that the two lines are very close to each other. Table II shows a sample of results achieved by the model where actual data vs. predicted data by the model is compared to each other.

B. Location-Wise Analysis

We have provided line graphs to visualize AQI data for various cities, with yellow bars indicating the minimum AQI, blue bars representing the average AQI, and red bars indicating

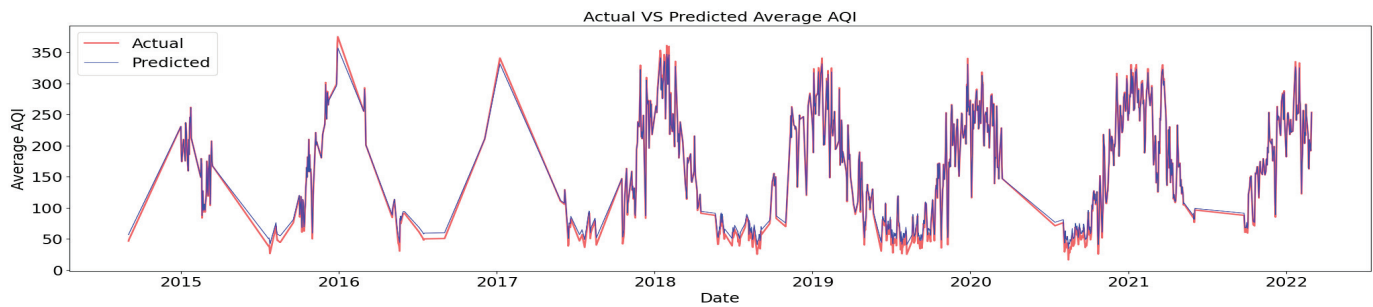


Fig. 2. Mean values for Actual VS Predicted AQI

the maximum AQI. Each figure also highlights the three time periods (before, during, and after COVID) with a black horizontal scale.

1) BARISHAL:

Located in the southern region of Bangladesh, this city has a relatively small population. Before the COVID-19 pandemic, the average AQI in this city ranged from 135 to 150. During the lockdown period, there were no significant changes in the AQI. However, after the lockdown was lifted, there was a slight increase in the average AQI to 170, as illustrated in Fig. 3.

2) CHITTAGONG:

The country's largest port city, situated on the Bay of Bengal, heavily relies on its core port. Despite the city's rapidly growing population, its geographical location has resulted in a lower AQI compared to Barishal. Before the COVID-19 pandemic, the city's average AQI remained below 130. During the lockdown, it dropped even further to below 100. However, after the lockdown was lifted, the average AQI rose to 130, as depicted in Fig. 4.

3) CUMILLA:

Cumilla has comparatively better air quality than other cities, despite the presence of the textile industry which emits various harmful chemicals and gases. The average AQI in Cumilla was recorded at 106 in 2019, which slightly increased to 118 during the lockdown and further rose to 160 after the lockdown as shown in Fig. 5.

4) DHAKA:

The largest capital city of Bangladesh, with a population of over 10 million people, experienced significant changes in its air quality index (AQI) during the COVID-19 pandemic. Before the pandemic, the average AQI ranged between 170-190, but during the lockdown period, the AQI reduced to 150. However, after the lockdown, the AQI drastically increased to 250, as depicted in Fig. 6.

5) GAZIPUR:

Gazipur is a city in Bangladesh that has a large number of garment factories, brick kilns, and highways, all of which contribute to high air pollution levels. Before COVID-19, the average AQI in Gazipur was around 170 and decreased slightly to 160 during the lockdown. However, the AQI sharply increased to 230 after the lockdown, as depicted in Fig. 7.

6) KHULNA:

The city of Mongla, the third largest in Bangladesh and home to the country's second busiest seaport experienced an

average AQI of around 120 before COVID-19. During the lockdown period, the AQI decreased to just under 100 but increased to above 180 after the lockdown, as indicated in Fig. 8.

7) MYMENSINGH:

Due to ongoing development projects and extensive air pollution caused by the Dhaka-Mymensingh highway, the city's AQI was around 200. Surprisingly, the average AQI increased from 140 to 190 during the lockdown period before jumping to 240 after the lockdown, as shown in Fig. 9.

8) NARSINGDI:

Our dataset for Narsingdi only contained AQI data from 2020, making it impossible to analyze AQI changes before COVID-19. However, during the lockdown period, the average AQI was recorded at 104, which increased to 150 after the lockdown, as shown in Fig. 10.

9) RAJSHAHI:

Rajshahi, a metropolitan city in the northern part of Bangladesh, had a high AQI of 500 in 2014, but due to a tree-planting campaign in 2016, the AQI decreased to 250-300 in the following years. During the lockdown period in 2020, the city's AQI declined, with an average of under 100 from March to December. However, the following year, the average AQI increased to above 200, as shown in Fig. 11.

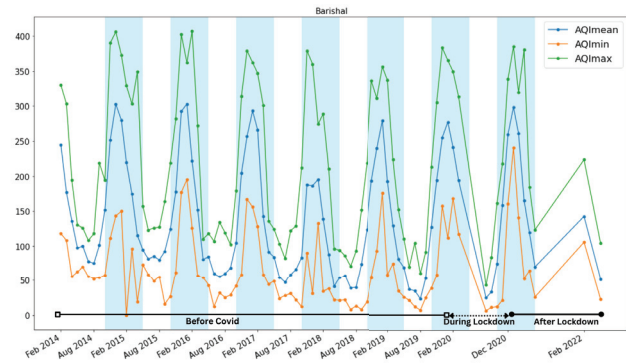


Fig. 3. Barishal AQI 2014-2022

10) SYLHET:

The city, famous for tea production, had an average AQI of 110 before the COVID-19 pandemic. However, due to the unavailability of 2020 data in our dataset, we could not determine the AQI changes during the lockdown period.

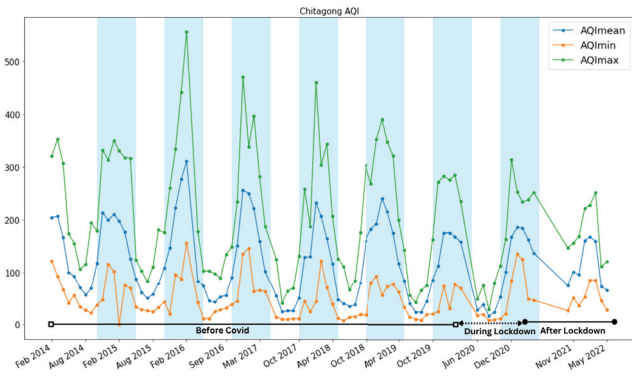


Fig. 4. Chittagong AQI 2014-2022

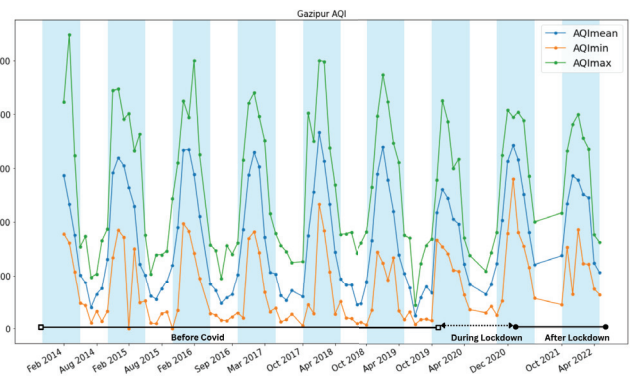


Fig. 7. Gazipur AQI 2014-2022

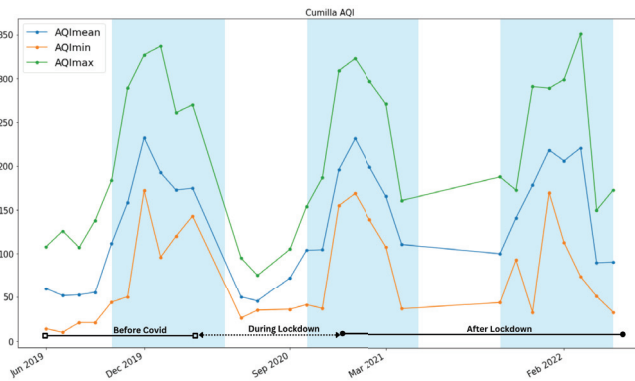


Fig. 5. Cumilla AQI 2014-2022

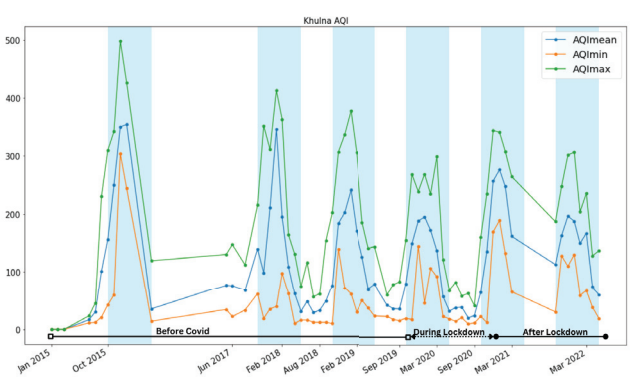


Fig. 8. Khulna AQI 2014-2022

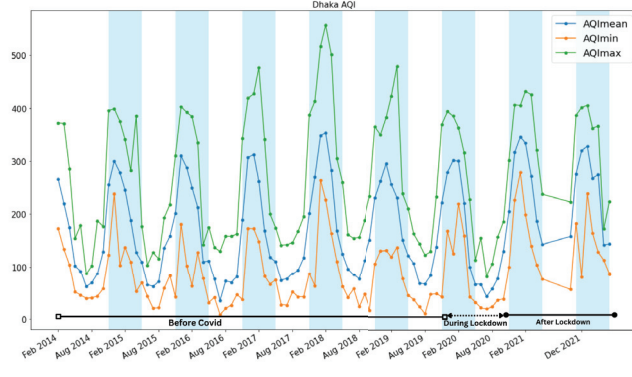


Fig. 6. Dhaka AQI 2014-2022

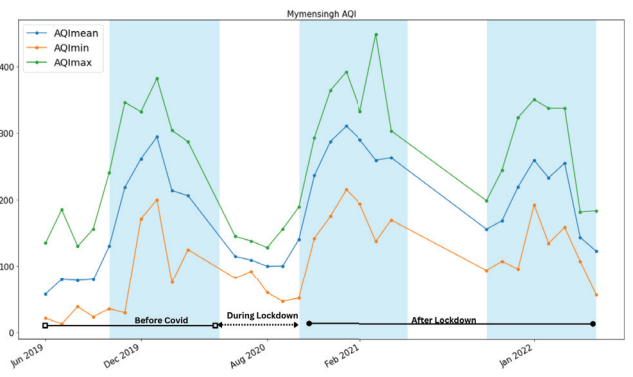


Fig. 9. Mymensingh AQI 2014-2022

Nonetheless, after the lockdown, the average AQI increased to 150, as indicated in Fig. 12.

C. Discussion

From our analysis, we found that the concentration of AQI is alarmingly high all around the nation, especially in major cities like Dhaka and commercial hubs like Chittagong and Gazipur. Dhaka is notorious for its pollution, and it has frequently been named the most polluted city and the city with the poorest Air Quality Index. Although the country's AQI is far worse than average, it was discovered that during the COVID lockdown, the AQI significantly decreased, as shown in the above figures. In comparison to the highest AQI reported in 2019 or 2021, the highest AQI in the nation during the

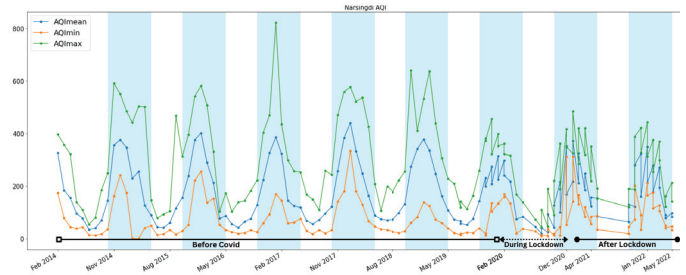


Fig. 10. Narsingdi AQI 2014-2022

lockdown period in 2020 was much lower. Additionally, we discovered that the lockdown period in 2020 had the lowest

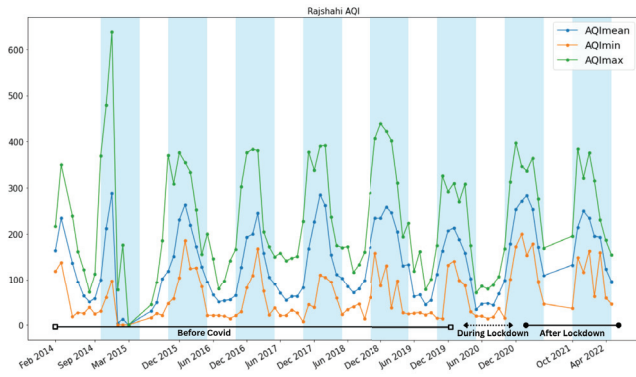


Fig. 11. Rajshahi AQI 2014-2022

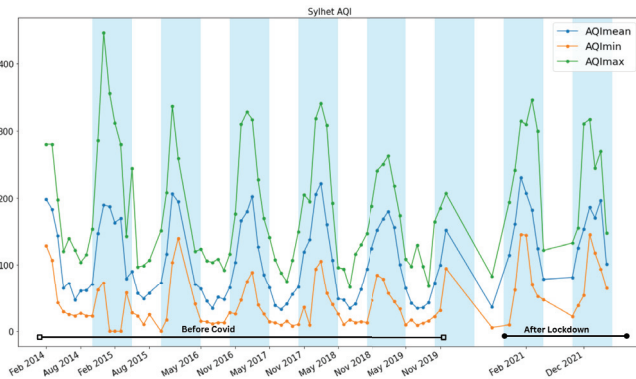


Fig. 12. Sylhet AQI 2014-2022

air quality readings compared to the lowest readings observed in 2019 and 2021. This shows that despite the country being shut down entirely for several weeks, air quality marginally improved during the COVID lockdown period in 2020. We also discovered a trend in the AQI; the country always had a higher AQI from December to March each year. These few months are the Dry season, which might cause the AQI to rise due to little to no rainfall. Although several of the main cities' average AQI were lower during the lockdown, it was also discovered—to no one's surprise—that some cities had high average AQI simultaneously. This was mostly caused by the fact that lockdown was not applied consistently across the nation. This was also noted when we discovered that the nation's average AQI in 2020 was only barely higher than the nation's average AQI in 2019.

VI. CONCLUSION

Bangladesh is known for its alarmingly high AQI concentrations due to massive air pollution. In this study, we aimed to evaluate the performance of a Support Vector Regression (SVR) model on AQI data and identify trends in AQI concentrations in Bangladesh. It has been identified that air pollution got reduced throughout the world during the COVID lockdown period. Even the neighboring countries had a relatively higher reduction in air pollution i.e., lower AQI concentrations compared to Bangladesh during the COVID lockdown. This demonstrates that, in contrast to other regions of the world, the general level of air pollution has only considerably decreased

as a result of poor lockdown implementation and that the AQI might have been significantly improved by properly implementing stringent lockdown procedures. We have also found SVR model predicts the AQI concentrations quite accurately with 6.2% MAPE. In the future, we plan to improve this model by incorporating deep learning techniques to predict future AQI concentrations. By doing so, we hope to contribute to mitigating the air pollution problem in Bangladesh.

REFERENCES

- [1] A. Benchrif, A. Wheida, M. Tahri, R. M. Shubbar, and B. Biswas, "Air quality during three covid-19 lockdown phases: Aqi, pm2. 5 and no2 assessment in cities with more than 1 million inhabitants," *Sustainable Cities and Society*, vol. 74, p. 103170, 2021.
- [2] G. Kyrkilis, A. Chaloulakou, and P. A. Kassomenos, "Development of an aggregate air quality index for an urban mediterranean agglomeration: Relation to potential health effects," *Environment international*, vol. 33, no. 5, pp. 670–676, 2007.
- [3] M. T. Latif, D. Dominick, N. S. S. L. Hawari, A. A. A. Mohtar, and M. Othman, "The concentration of major air pollutants during the movement control order due to the covid-19 pandemic in the klang valley, malaysia," *Sustainable Cities and Society*, vol. 66, p. 102660, 2021.
- [4] M. S. U. Miah, T. B. Sarwar, S. S. Islam, M. S. Haque, M. Masuduzzaman, and A. Bhowmik, "An adaptive medical cyber-physical system for post diagnosis patient care using cloud computing and machine learning approach," in *2022 3rd International Conference for Emerging Technology (INCET)*, 2022, pp. 1–6.
- [5] M. Mofijur, I. Rizwanul Fattah, A. Saiful Islam, M. Uddin, S. Ashrafur Rahman, M. Chowdhury, M. A. Alam, and M. A. Uddin, "Relationship between weather variables and new daily covid-19 cases in dhaka, bangladesh," *Sustainability*, vol. 12, no. 20, p. 8319, 2020.
- [6] "Bangladesh Air Quality Index (AQI) and Air Pollution information | IQAir." [Online]. Available: <https://www.iqair.com/bangladesh>
- [7] M. M. Rahman, S. Mahamud, and G. D. Thurston, "Recent spatial gradients and time trends in dhaka, bangladesh, air pollution and their human health implications," *Journal of the Air & Waste Management Association*, vol. 69, no. 4, pp. 478–501, 2019.
- [8] "bangladesh-air-quality/data/silver/case at master · TSGreen/bangladesh-air-quality." [Online]. Available: <https://github.com/TSGreen/bangladesh-air-quality>
- [9] R. P. Singh and A. Chauhan, "Impact of lockdown on air quality in india during covid-19 pandemic," *Air Quality, Atmosphere & Health*, vol. 13, no. 8, pp. 921–928, 2020.
- [10] M. Islam, T. A. Chowdhury *et al.*, "Effect of covid-19 pandemic-induced lockdown (general holiday) on air quality of dhaka city," *Environmental Monitoring and Assessment*, vol. 193, no. 6, pp. 1–16, 2021.
- [11] M. S. Rahman, M. A. K. Azad, M. Hasanuzzaman, R. Salam, A. R. M. T. Islam, M. M. Rahman, and M. M. M. Hoque, "How air quality and covid-19 transmission change under different lockdown scenarios? a case from dhaka city, bangladesh," *Science of The Total Environment*, vol. 762, p. 143161, 2021.
- [12] M. Masum and S. Pal, "Statistical evaluation of selected air quality parameters influenced by covid-19 lockdown," *Global Journal of Environmental Science and Management*, vol. 6, no. Special Issue (Covid-19), pp. 85–94, 2020.
- [13] Z. Qiu, M. A. Ali, J. E. Nichol, M. Bilal, P. Tiwari, B. A. Habtemicheal, M. Almazroui, S. K. Mondal, U. Mazhar, Y. Wang *et al.*, "Spatiotemporal investigations of multi-sensor air pollution data over bangladesh during covid-19 lockdown," *Remote Sensing*, vol. 13, no. 5, p. 877, 2021.
- [14] "Time series forecasting with support vector regressor." [Online]. Available: <https://github.com/microsoft/ML-For-Beginners/tree/main/7-TimeSeries>
- [15] S. S. Islam, M. S. Haque, M. S. U. Miah, T. B. Sarwar, and R. Nugraha, "Application of machine learning algorithms to predict the thyroid disease risk: an experimental comparative study," *PeerJ Computer Science*, vol. 8, p. e898, 2022.
- [16] M. J. Hossain, S. M. Abdullah, M. Barkatullah, M. S. U. Miah, T. B. Sarwar, and M. F. Monir, "Predicting the success of suicide terrorist attacks using different machine learning algorithms," in *2022 25th International Conference on Computer and Information Technology (ICCIT)*, 2022, pp. 378–383.